

NUMBER CRUNCHING IN THE BIST

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Who is this study aimed at?

- Let's assume you have gone back in time to 12th June, 2012.
- The BIST100 index posted a **positive return in each of the last 8 consecutive days**, as shown on the right
- You are about to decide whether to take a position, and in which direction.
- You've analyzed the current market environment, assessed potential market moving events and analysed market fundamentals and technicals.
- But you also realized that having positive returns for 8 consecutive days is **not a very common event**.
- Before deciding on your trade, do you think a knowledge of **how the BIST100 index behaved in similar situations in the past** would help your decision?
- If **'YES'**, then you may be interested in the findings of this study.



What is this study about?

- This study includes some **interesting statistics** about the BIST100 index
- We used **24 years of data** to analyse historical patterns in the index
- The study has **4 sections**:
 - ✓ Some descriptive characteristics of the BIST100 index: Typical return profiles over certain time periods (e.g. daily, weekly), volume and volatility patterns
 - ✓ Calendar effects: How the index returns differ within different months of the year, different days of the month, different days of the week, etc
 - ✓ Major drawdowns: Major corrections in the history of BIST100 index
 - ✓ Event profitability: The profitability of investing in the BIST100 index after the occurrence of certain events, such as moving average crossovers, major drawdowns or volume breakouts

How is this study useful?

- As a **measurement tool**:
 - ✓ This study includes different statistics calculated using past data.
 - ✓ These statistics can be used as a measurement tool to assess **how extreme a recent or a potential future event is in a historical context**.
 - ✓ For instance, knowing that the BIST100 index is currently experiencing its sixth worst drawdown in its history may help investors assess the magnitude of the current event.
 - ✓ Accordingly, these statistics may be used as a supporting argument in forming market views
- As a tool to assess **event profitability**:
 - ✓ The last section of the study includes the historical profitability metrics for various events
 - ✓ These statistics may be used to assess the **potential profitability** of short term trades based on those events

What is not covered in this study?

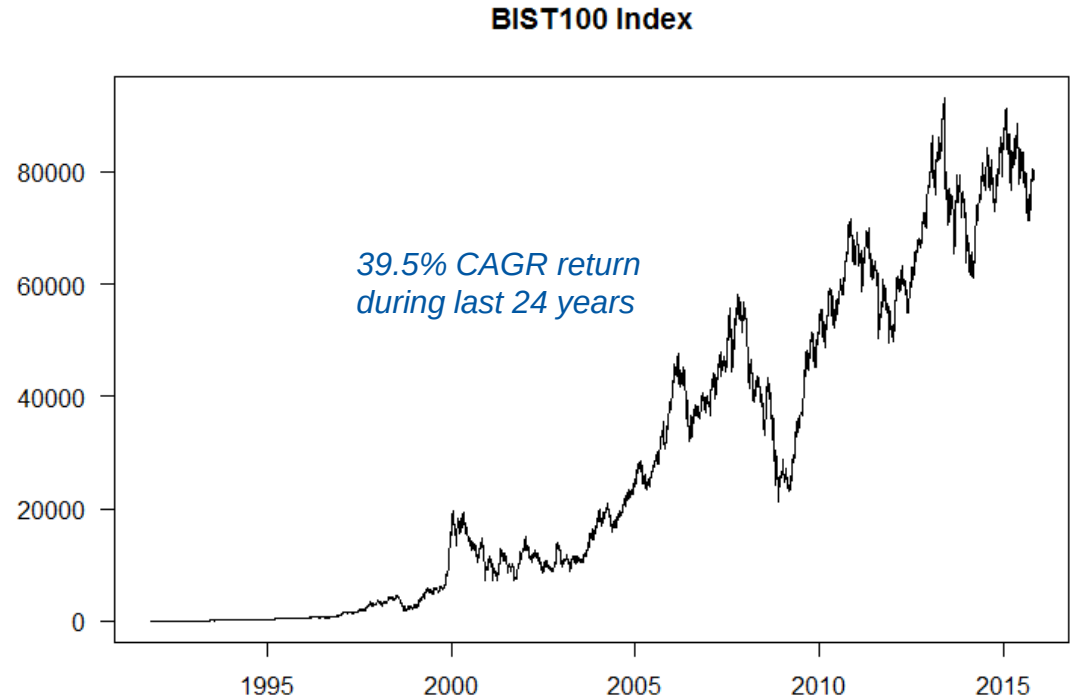
- All statistics reported here are calculated using historical data and thus are implicitly based on a 'history repeats itself' argument. However, future realizations may potentially differ significantly from historical events, and there always exists the possibility of seeing 'black swan' type events in the future.
- The analysis of event profitability include profitability metrics for fixed investment horizons. Therefore, systematic trading based on certain events using various buy/sell signals are out of scope of this study.
 - ✓ For example, we analyzed the profitability of investing in BIST100 index when 20 day moving average crosses above 50 day moving average and holding our position for a fixed 1 week investment period.
 - ✓ On the other hand, a systematic trading strategy may take a long position when the 20 day moving average crosses above 50 day moving average and close the position when 20 day moving average crosses below 50 day moving average. The profitability of such a strategy with variable investment horizons is out of scope of this study.
- The profitability statistics presented here do not include any transaction fees

MEET THE BIST100 INDEX

Some Descriptive Characteristics

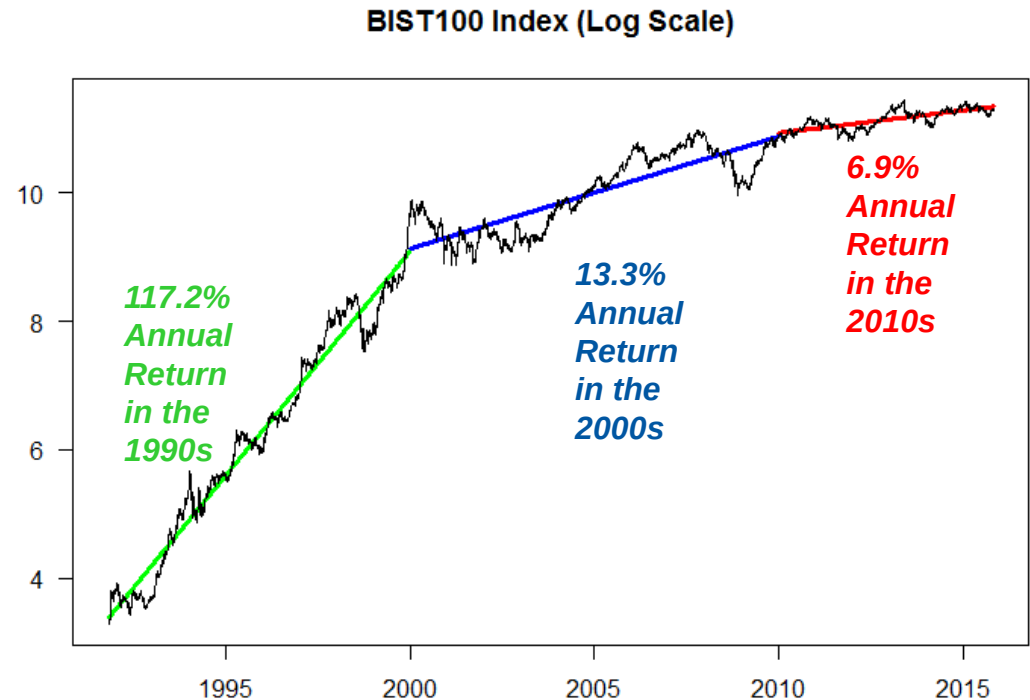
Meet the BIST100 index

- If you had invested **1 TRY** in BIST100 index 24 years ago, your investment would grow to **2948 TRY** today
- The index earned **39.5%** average annual return during this period
- However the road to such a high return was not a smooth one, with many corrections and bear market periods.
- The index spent **38% of its time in a correction** (defined as periods with a drawdown of more than 10%), and 62% of the time in bull periods.



With three different growth episodes

- The growth rate differed significantly in each decade
- High inflation periods typically coincided with high growth periods in BIST
- The average annual return in 1990s was an eye-catching **117.2%**.
- This declined to **13.3%** in the 2000s and a mere **6.9%** in the 2010s.

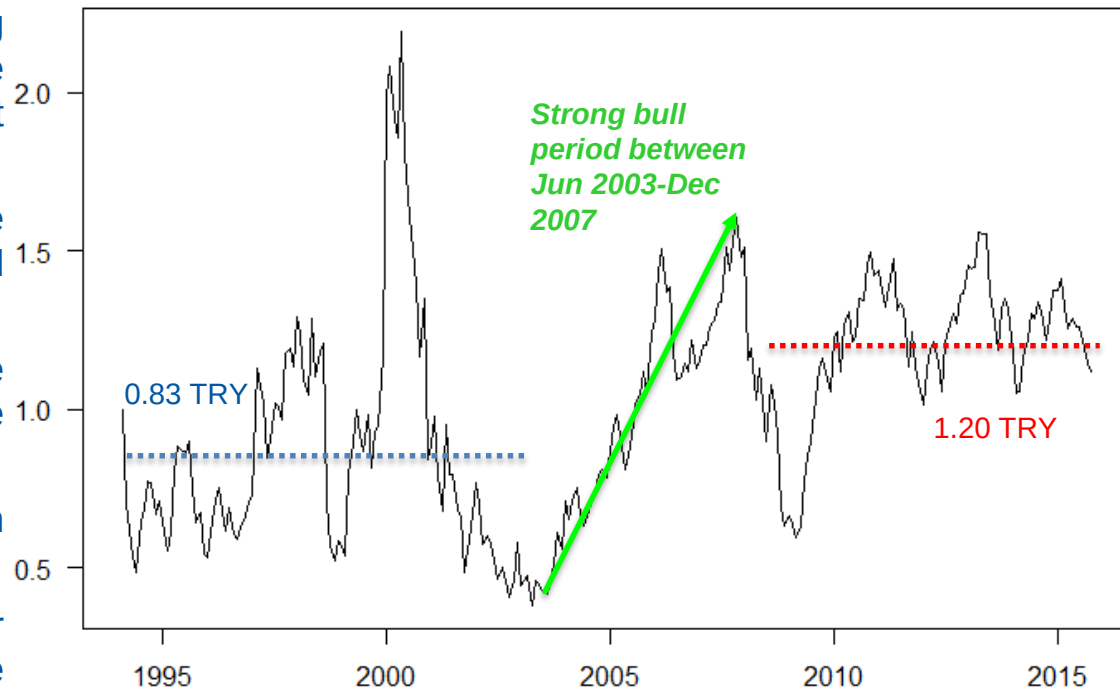


Not a growth story in inflation-adjusted real terms

- Despite high nominal returns during the last 24 years, the real value of the index adjusted for inflation was not actually a growth story
- In real terms, the performance of the BIST100 index can be characterized in 3 different periods:

- ✓ **Range bound** behaviour in the 1990s and until 2003 (the average real value was 0.83 TRY)
- ✓ **A strong bull** period between 2003-2007
- ✓ **Range bound** behaviour afterwards (the average real value was 1.20 TRY)

Inflation Adjusted Value of BIST100 Index

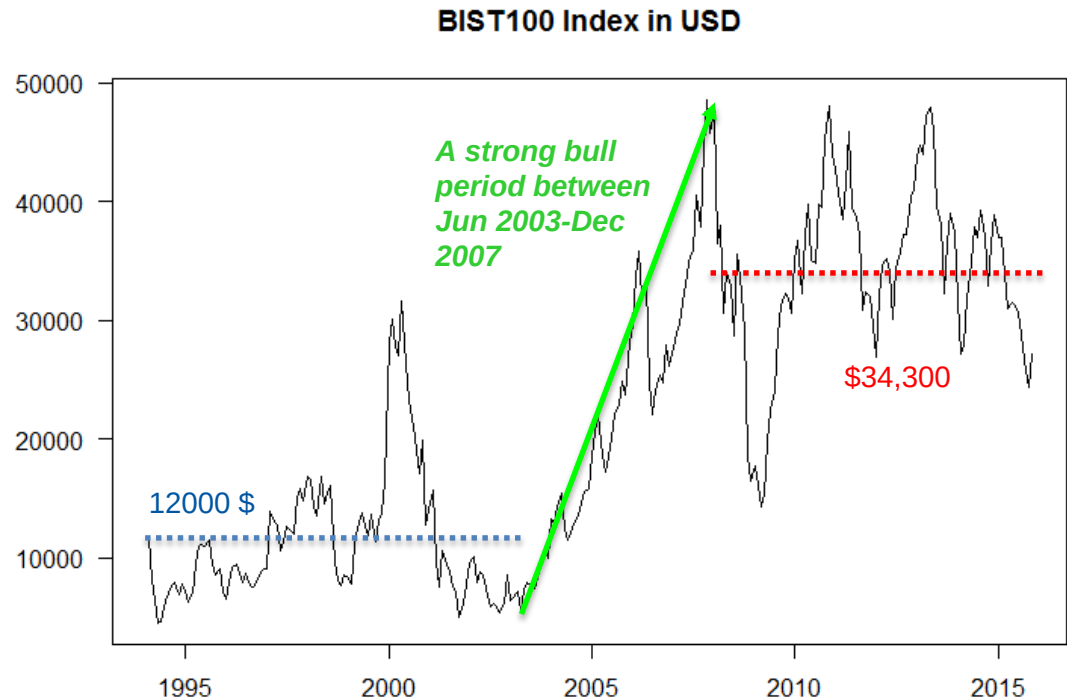


Note: BIST100 divided by consumer price index. Rebased to 1 TRY at inception

- The strong bull period between 2003-2007 coincides with improved macro fundamentals such as lower inflation and interest rates, a slimmer budget deficit and a reduced debt burden.
- This implies that **only the structural transformations in Turkish economy** yielded a real appreciation in the BIST, while cyclical improvements in the economic outlook only give rise to short term rallies in real terms

Nor in USD terms

- The BIST100 index presents a similar picture with its real value in USD terms
- Apart from the **strong appreciation** in USD terms between 2003-2007, the index followed **range bound** behaviour in the 1990s and after 2007



Patience is a virtue in investing

- 51.5% of daily returns were positive, while 62.5% of all calendar years had positive returns.
- On a rolling basis, 75.3% of all possible one-year periods (approximately 260 business days) starting from different days yielded a positive return
- These figures confirm that investing in longer terms lowers the possibility of a loss

Calendar Returns

Investment Horizon	% with a negative Return	% with a Zero Return	% of Positive Return
Daily Returns	45.1%	3.4%	51.5%
Weekly Returns	42.6%	1.7%	55.7%
Monthly Returns	41.0%	0.3%	58.7%
Yearly Returns	37.5%	0.0%	62.5%

Rolling Returns in Overlapping Periods

Investment Horizon	% of Negative Return	% of Zero Return	% of Positive Return
1 Business Day	45.1%	3.4%	51.5%
5 Business Days	43.1%	1.1%	55.8%
20 Business Days	42.0%	0.5%	57.5%
260 Business Days	24.5%	0.2%	75.3%

What return can be expected in a typical day, week, month or year?

- What return would you expect to earn in a typical day? What about a typical weekly return?
- Historical statistics show that a typical daily return is 0.1%, or 0.7% for a typical week, **2.1% for a typical month** and 34.6% for a typical year.
- In all of the analysed investment horizons, typical returns for bullish periods are higher (in absolute terms) than typical returns for bearish periods. For example, for monthly investment horizon (approximately 20 business days), a typical positive return is 8.3%, while a typical negative return is -5.9%.
- The worst return over any one year investment horizon (approximately 260 business days) was **-60.5%**, while the index earned a **696.3%** return in the best year

Return Statistics

Investment Horizon	Minimum Return	Median Negative Return	Median Return	Median Positive Return	Maximum Return
1 Business Day	-18.1%	-1.3%	0.1%	1.4%	19.5%
5 Business Days	-30.8%	-3.0%	0.7%	3.4%	41.6%
20 Business Days	-51.6%	-5.9%	2.1%	8.3%	94.3%
260 Business Days	-60.5%	-15.8%	34.6%	53.8%	696.3%

How long can a bullish/bearish period last?

- Having 10 consecutive weeks with positive returns is not so common in BIST100 index. Nevertheless it occurred 2 times in its history.
- Similarly the index experienced 5 different periods when it had daily losses in 8 consecutive days
- The tables on the right present the historical frequency of observing positive or negative returns in different consecutive investment periods
- These statistics can be used to assess how long a bullish or bearish period can last.
- For instance the index had no 7 consecutive weeks with negative returns. So, if you observed negative returns in the last 6 weeks, then you would have a good reason to turn more bullish for the next week.

BIST100 index had 10 consecutive weeks with positive returns 2 times in its history

Number of cases with consecutive N positive returns

N	Daily	Weekly	Monthly	Yearly
12	0	1	0	0
11	1	2	0	0
10	11	2	0	0
9	15	4	1	0
8	22	7	3	0
7	30	11	5	0
6	62	18	6	0
5	112	31	10	1

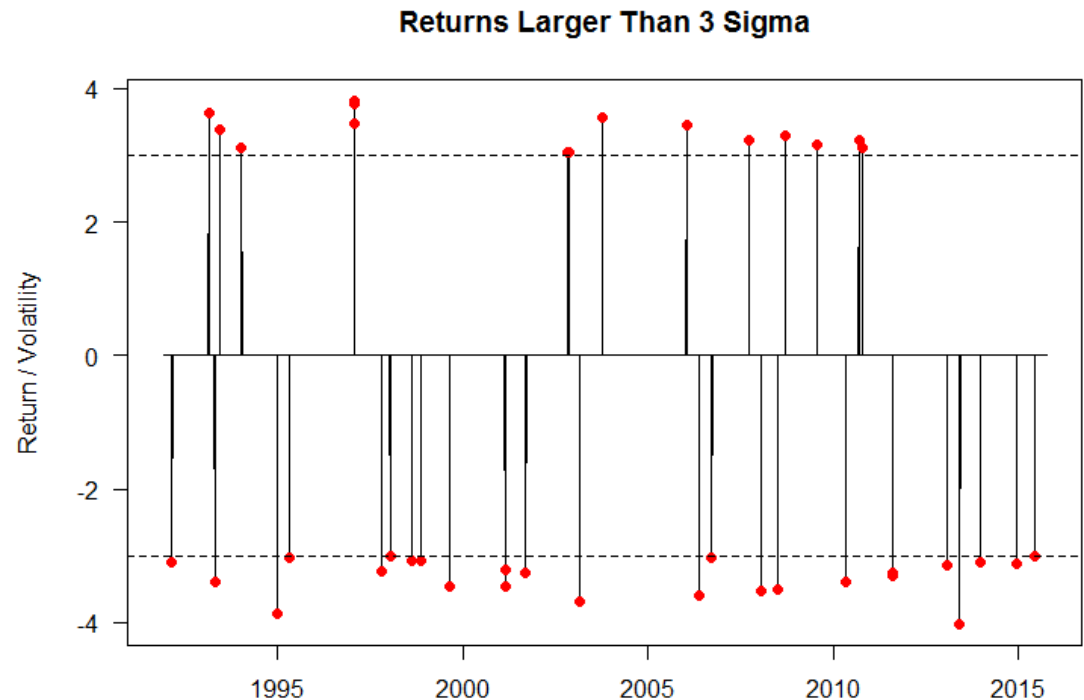
Number of cases with consecutive N negative returns

N	Daily	Weekly	Monthly	Yearly
11	2	0	0	0
10	2	0	0	0
9	2	0	0	0
8	5	0	0	0
7	16	0	0	0
6	31	3	0	0
5	60	8	2	0

BIST100 index had 8 consecutive days with negative returns 5 times in its history

Extreme returns are more frequent than you think

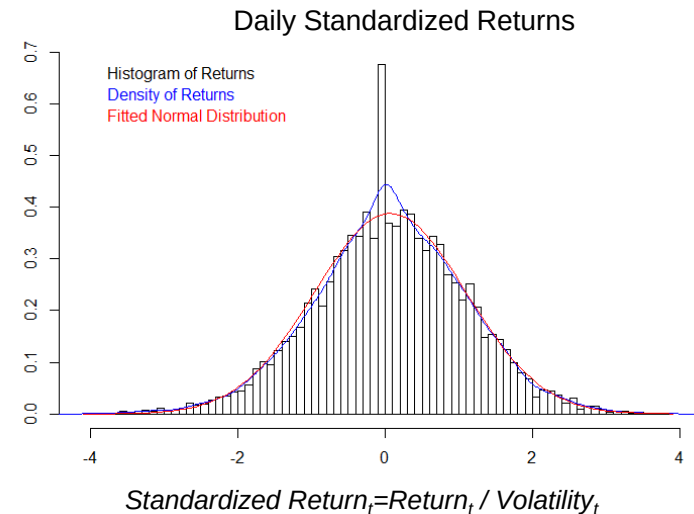
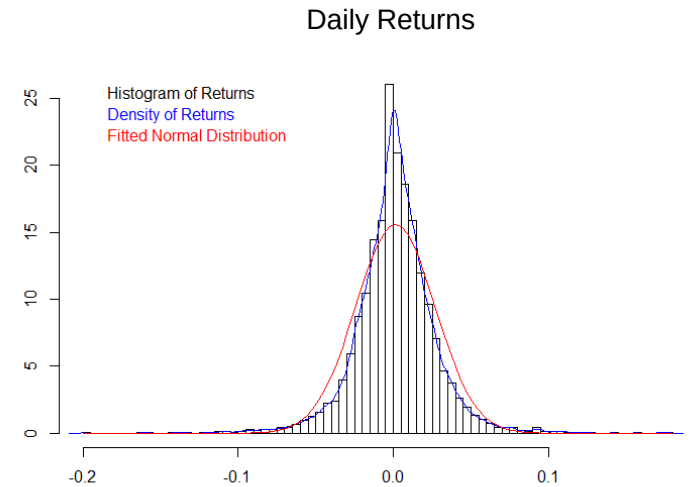
- Extreme returns are defined as the ones that are larger than 3 times the volatility for that day, sometimes called as '3+ sigma events'
- Extreme returns occurred in 40 different days throughout 24 years. Thus you may expect to see one in every 1.5 years on average
- If happen, extreme returns most probably have a negative sign. In the past, 25 out of 40 extreme returns had negative sign



Note: Volatility is calculated as the standard deviation of returns in past 20 business days

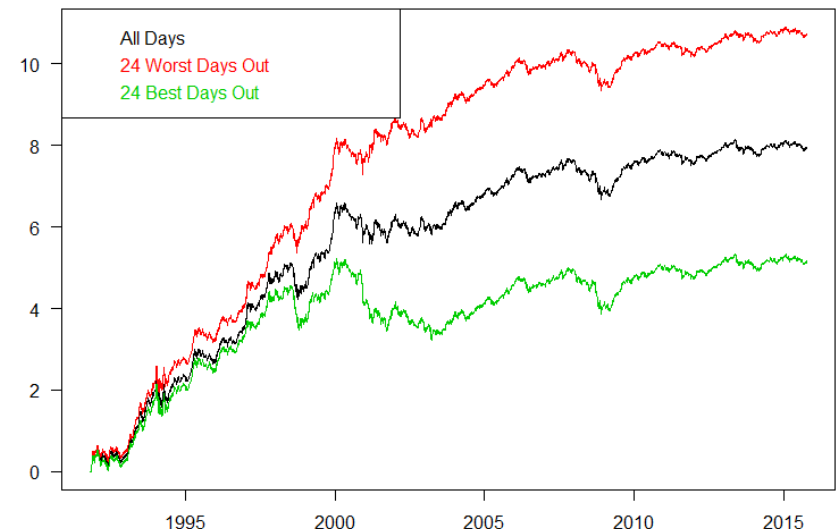
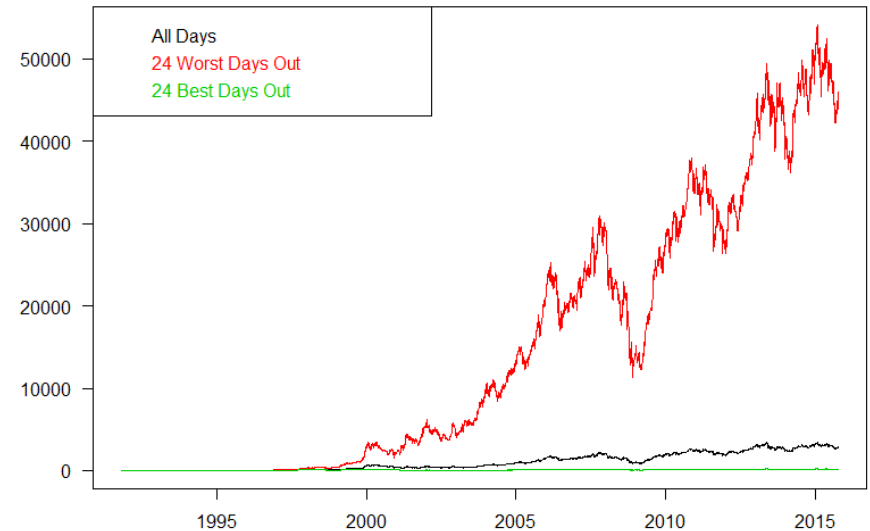
The famous bell curve: Not for returns, unless you standardize them

- Daily returns do not follow a normal distribution (i.e. the famous bell curve).
- Returns typically have more frequent large positive or negative returns than what normal distribution suggests. This means that the observed return distribution has 'fatter tails' than the normal distribution.
- But once we standardize the returns by dividing them by the daily volatility, then the resulting distribution seems more like normal distribution



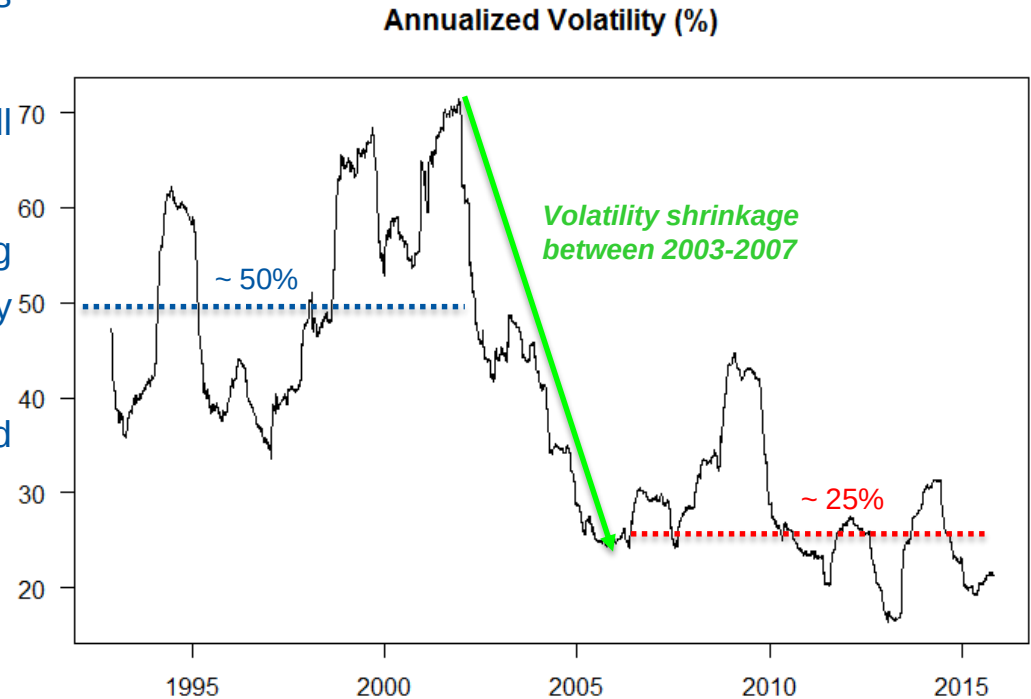
The cost of bad market timing: forget 40% of your CAGR return

- Missing a small number of critical days may significantly change the resulting return that you get
- Throughout the last 24 years, missing the 24 best days would reduce your average annual return from 39.5% to 24.0%, which corresponds to a 40% reduction in annualized terms.
- Similarly, if you managed to avoid the worst 24 days in 24 years, you would end up with an average annual return of 56.6%, corresponding to a 43% increase from a buy and hold investment.
- Thus cost of bad market timing or, conversely, the benefit from good market timing, may be as high as 40% of your annual return
- The resulting investment returns are shown in the graphs on the right hand side. The first graph has a nominal scale while the second is logarithmic.



Lower volatility in the BIST

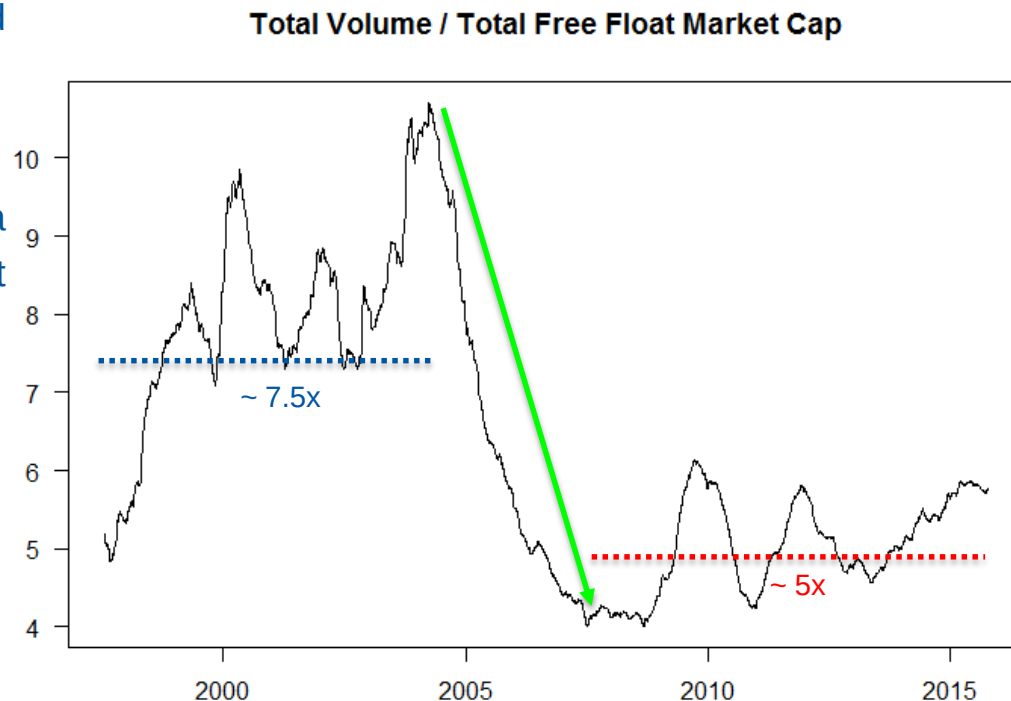
- The volatility of the BIST100 index has **almost halved** over the last 3 decades
- It was around **50%** during 1990 and until 2003
- Improved macro fundamentals during 2003-2007 period caused a volatility shrinkage
- Since 2007, it has hovered at around **25%**



Note: Volatility is calculated as the standard deviation of daily returns for the last 260 days, and annualized

BIST Volume

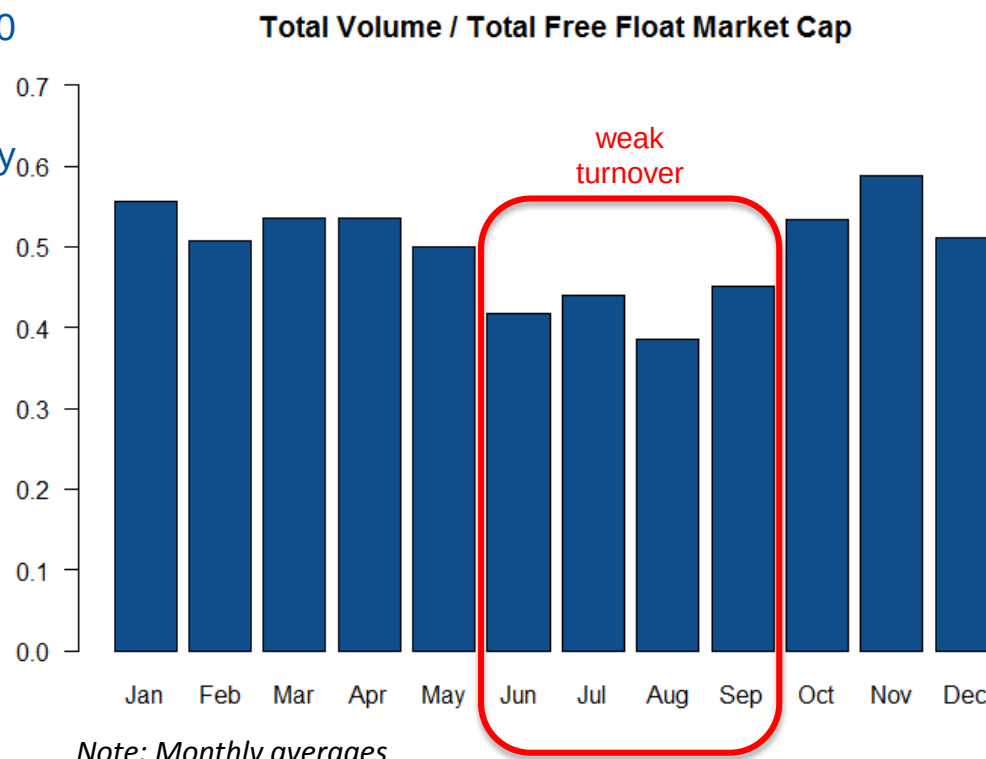
- Annual turnover on the BIST100 hovered at around **7.5x** before 2003
- After 2007, it declined to **5x**
- A 5x turnover per year corresponds to a weekly two-way transaction volume that is equal to **10% of the portfolio value**



Note: Calculated as one year rolling average

The BIST goes on holiday in the summer, like everyone else

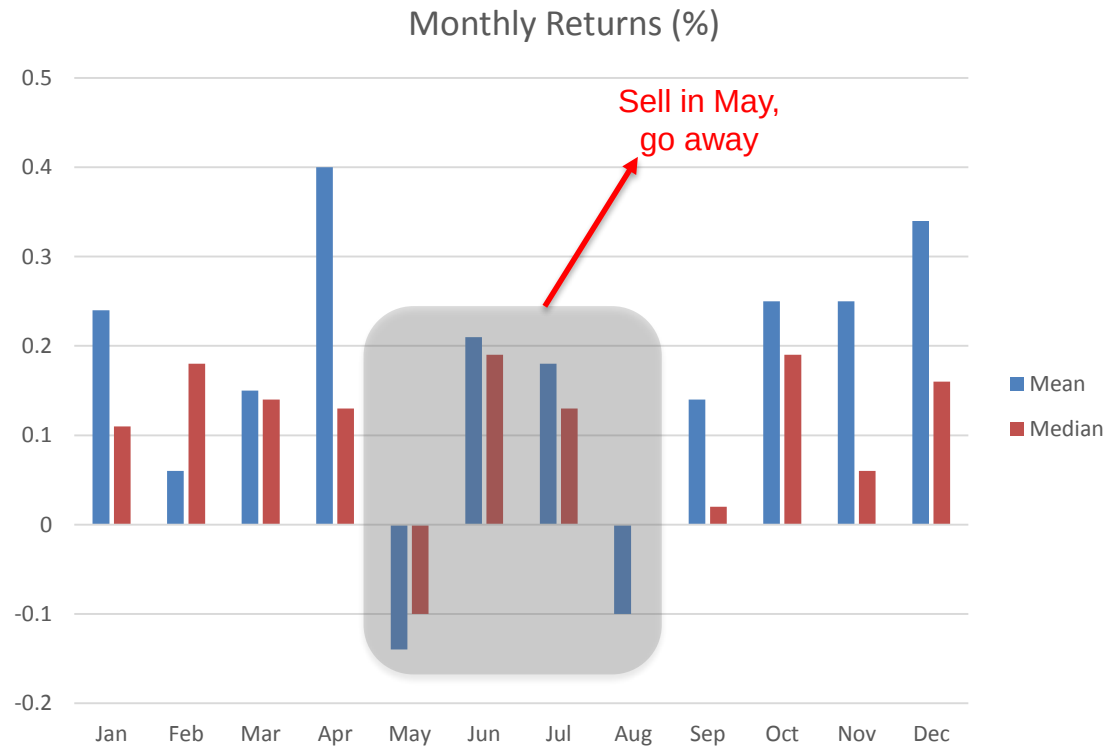
- Average monthly turnover in the BIST100 is weak during the **summer**
- The highest average turnover is typically seen in **November**



CALENDAR EFFECTS

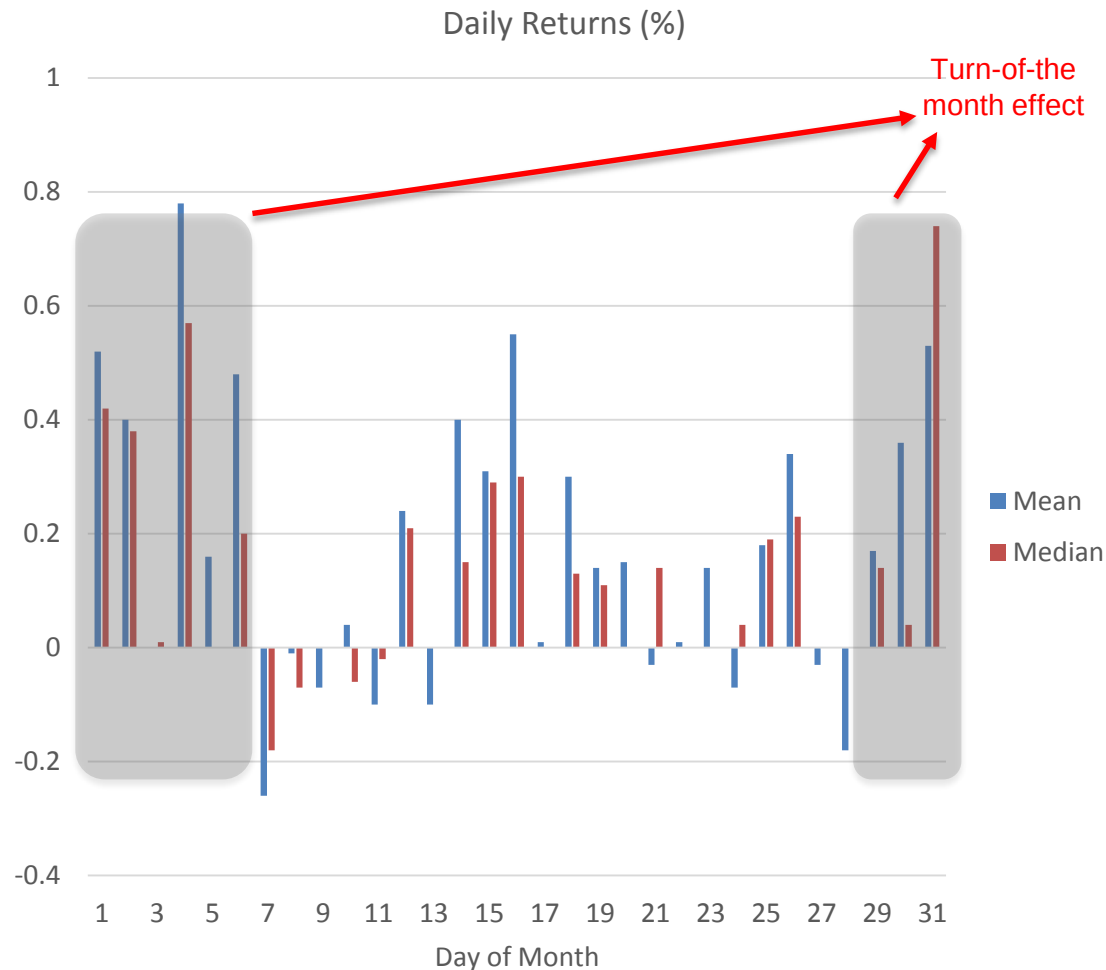
Month of the year ; No surprise

- Not all months of the year appear to be same for the BIST
- Typical monthly returns are negative in May and August, but positive for other months
- There does indeed seem to be a 'sell-in-May, go away' phenomenon in the BIST100 index.



Day of the month; Trade at the beginning or the end...

- Different days of the month also provide quite different returns
- Most positive returns in the BIST100 index occurred on the very first and last days of the month
- Thus there seems to be a turn-of-the-month effect in the BIST100 index



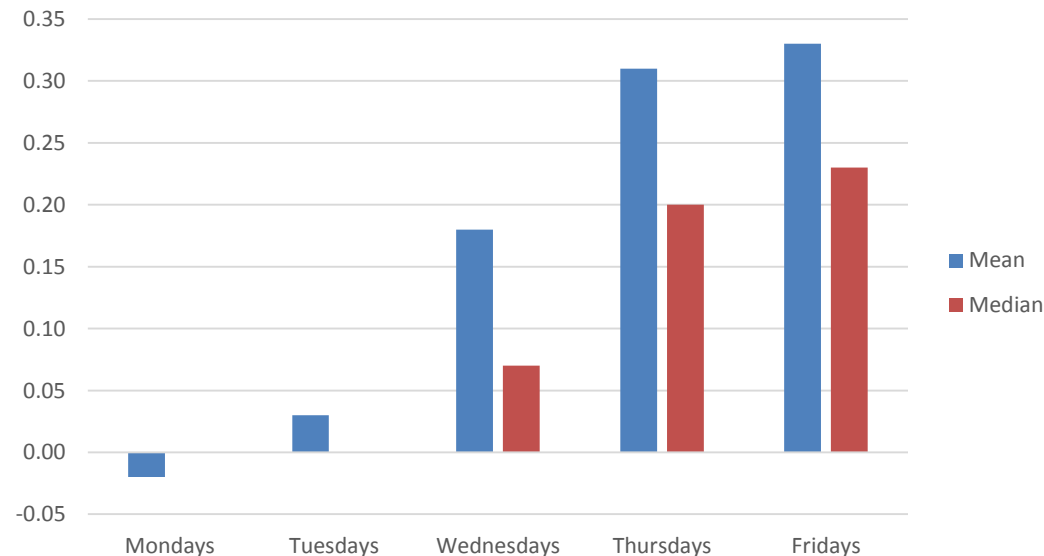
Day of the week; Happy Friday !

- Typical returns improve as we go from Mondays to Fridays
- Mondays and Tuesdays had zero median returns, while a typical Friday investment yielded 0.23% daily return

Daily Returns (%)

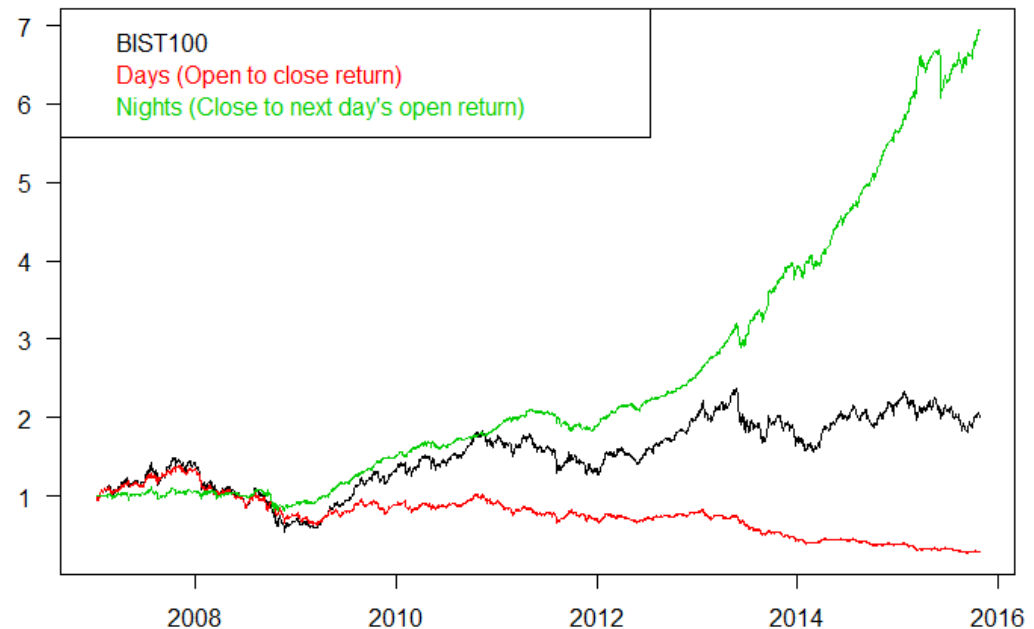
	Minimum	Mean	Median	Maximum
Mondays	-14.62	-0.02	0.00	13.11
Tuesdays	-9.46	0.03	0.00	19.45
Wednesdays	-18.11	0.18	0.07	18.64
Thursdays	-13.12	0.31	0.20	12.52
Fridays	-11.33	0.33	0.23	16.93

Daily Returns (%)



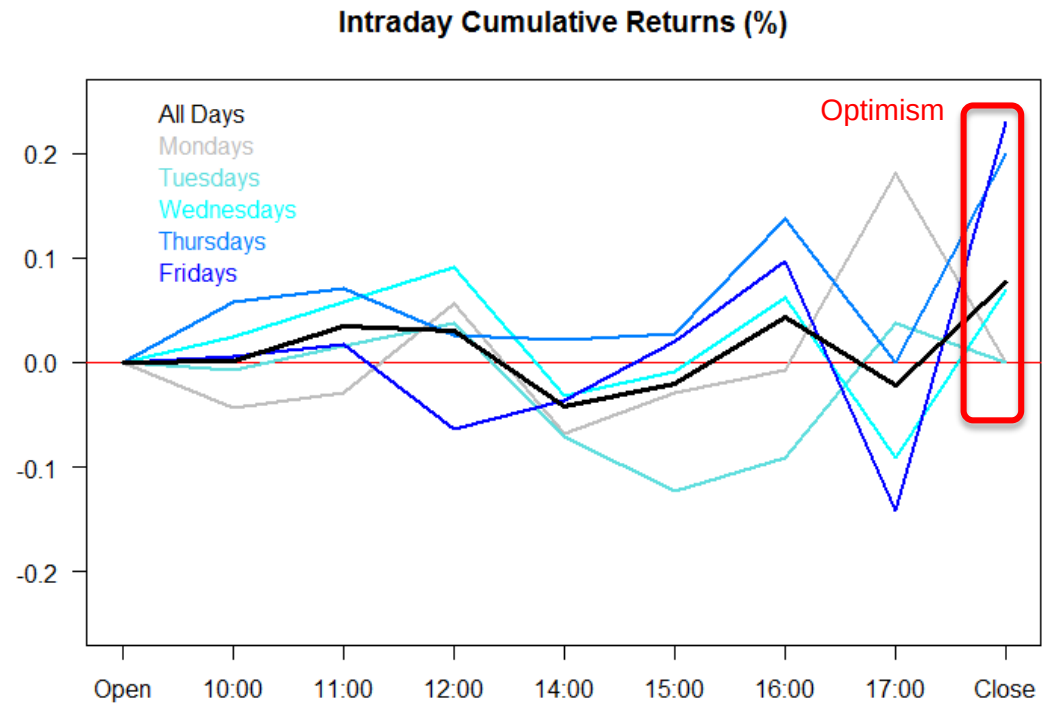
Days vs nights; Victory of the night crawlers

- During the last decade, almost all of the index return was earned at **night** when the market was closed.
- Indeed, an investor who only invested in **day returns** (as defined as the return from daily opening to the close) would have **lost money**



Hour of the day

- Intraday returns are not uniform throughout the trading hours
- **Optimism** appears more pronounced towards the **closing hours**



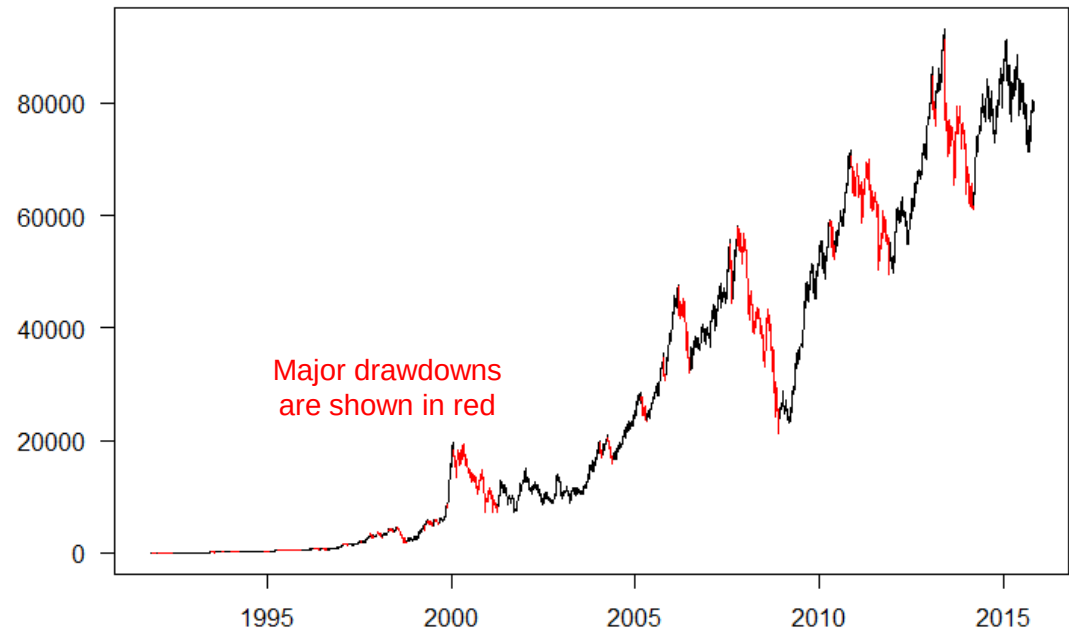
Note: Median cumulative returns per intraday hours

MAJOR DRAWDOWNS IN BIST

Major drawdowns

- We define major drawdowns as corrections of **10% or more** in the BIST100 index
- A total of **31** major corrections have occurred in BIST – one in every **1.3 years**
- In a typical major drawdown, the BIST100 index lost **19.4%** of its value within 27 business days, and took 35 business days for the index to recover its losses
- Thus, on average, it takes **50% more time** to recover from a major drawdown

Main Drawdowns in BIST100



	Best	Median	Worst
Percent Loss	-10.0%	-19.4%	-63.6%
Peak-to-through duration	4 business days	27 business days	294 business days
Through-to-new high duration	3 business days	35 business days	690 business days
Total duration	8 business days	61 business days	984 business days

The worst major drawdowns coincide with crises

2008 Lehman collapse
2000-2001 Banking crisis (TR)
1998 Russian crisis
1994 Currency crisis (TR)
1992 EU currency crisis

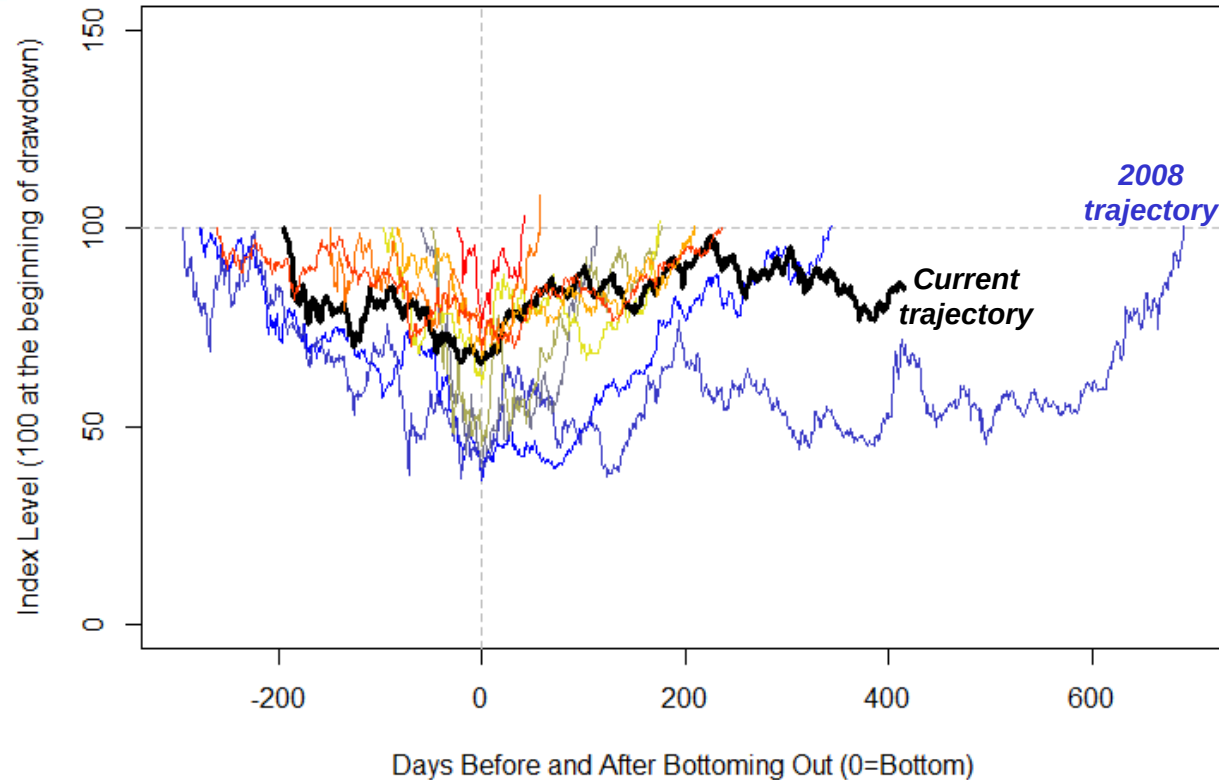
2006 Global financial turmoil

Begin Date	Trough Date	End Date	Max Loss	Total Duration (business days)	Peak-to-Trough Duration (business days)	Trough-to-New High Duration (business days)
16/10/2007	20/11/2008	05/04/2010	-63.5%	621	278	343
19/01/2000	29/03/2001	05/01/2004	-63.4%	984	294	690
17/07/1998	08/10/1998	26/03/1999	-59.1%	173	60	113
14/01/1994	24/03/1994	02/12/1994	-55.0%	225	49	176
17/01/1992	03/06/1992	12/02/1993	-39.2%	271	96	175
23/05/2013	03/03/2014	(To Date)	-34.3%	610	194	416 (No recovery yet)
28/02/2006	26/06/2006	25/04/2007	-33.1%	293	84	209
24/04/1995	29/11/1995	26/02/1996	-30.9%	207	150	57
10/11/2010	24/11/2011	30/10/2012	-30.6%	496	261	235
06/01/1998	10/02/1998	15/04/1998	-25.2%	65	24	41
24/10/1997	14/11/1997	05/01/1998	-24.0%	49	15	34
30/03/2004	17/05/2004	07/09/2004	-23.8%	113	34	79
05/07/1993	27/07/1993	27/08/1993	-22.9%	40	17	23
24/07/2007	16/08/2007	05/10/2007	-20.0%	53	18	35
12/05/1999	29/06/1999	19/07/1999	-19.5%	48	34	14
09/04/1996	28/05/1996	03/07/1996	-19.4%	58	32	26
28/01/1997	05/03/1997	23/06/1997	-19.3%	98	25	73
08/04/1999	13/04/1999	20/04/1999	-19.1%	9	4	5
01/03/2005	18/04/2005	14/07/2005	-18.0%	97	35	62
24/04/1998	01/06/1998	03/07/1998	-17.4%	50	26	24
22/07/1999	31/08/1999	08/09/1999	-16.4%	27	21	6
05/10/1993	02/11/1993	19/11/1993	-16.0%	32	19	13
27/11/1991	18/12/1991	23/12/1991	-15.6%	19	16	3
05/12/1994	23/01/1995	22/02/1995	-15.5%	58	36	22
09/07/1996	16/08/1996	02/10/1996	-15.1%	61	29	32
12/01/2004	06/02/2004	19/03/2004	-14.9%	46	16	30
05/10/2005	19/10/2005	21/11/2005	-13.6%	32	11	21
25/01/2013	22/02/2013	02/05/2013	-12.2%	68	21	47
15/04/2010	25/05/2010	21/07/2010	-11.9%	68	27	41
15/07/1997	28/07/1997	07/08/1997	-10.1%	18	10	8
23/11/1999	26/11/1999	02/12/1999	-10.0%	8	4	4

- The BIST100 index recently suffered its sixth largest drawdown, and has still not recovered
- Drawdowns that are more severe than the current one have always coincided with a major global or local crisis

BIST100 Index trajectory in the worst ten drawdowns

- The current drawdown is the **second worse** in history (after the one in 2008 when Lehman collapsed) in terms of the length of the recovery



Note: Each of the ten worst drawdowns are shown in different colours. The BIST100 Index level is rebased to 100 at the beginning of the drawdown. The x-axis shows the time and is rebased to zero when the index has bottomed out in each drawdown period

Major drawdowns in USD terms

- In USD terms, the current drawdown is also the **sixth largest** in history with a **54%** maximum loss
- For the current drawdown, the largest loss (i.e. trough) happened only recently on **11 September 2015**

Begin Date	Trough Date	End Date	Max Loss	Total Duration (business days)	Peak-to-Trough Duration (business days)	Trough-to-New High Duration (business days)
19/01/2000	09/10/2001	27/02/2006	-86.8%	1521	429	1092
14/01/1994	04/05/1994	23/10/1997	-78.7%	946	77	869
16/10/2007	20/11/2008	12/10/2010	-74.2%	752	278	474
24/10/1997	07/10/1998	09/12/1999	-65.5%	519	237	282
17/01/1992	18/11/1992	14/06/1993	-54.7%	345	209	136
10/11/2010	11/09/2015	(To Date)	-54.0%	1248	1218	30 (No recovery yet)
02/03/2006	26/06/2006	02/07/2007	-48.1%	339	82	257
05/07/1993	27/07/1993	08/09/1993	-26.1%	47	17	30
26/07/2007	20/08/2007	27/09/2007	-25.0%	45	18	27
05/10/1993	03/11/1993	29/11/1993	-21.6%	38	20	18
27/11/1991	18/12/1991	16/01/1992	-17.6%	36	16	20
01/12/1993	13/12/1993	22/12/1993	-10.1%	16	9	7

EVENT PROFITABILITY

Event profitability analysis

- This section includes the **historical profitability metrics** for investing in the BIST100 index after the occurrence of certain events, such as moving average crossovers, major drawdowns and volume breakouts
- We assume that we took a **long position in the BIST100** index at the end of the day on which a specific event occurred, and held our position for a fixed investment horizon
- We employ six different **investment horizons**: 1 day, 3 days, 1 week, 2 weeks, 1 month and 3 months
- These statistics may be used to assess the **potential profitability** of short term trades based on those events
- The profitability statistics presented here do not include any **transaction fees**

How to interpret the tables in this section

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
Daily return > 5%	47	Median Return (%)	0.36	1.72	2.06	2.59	2.50	5.26
		Hit Ratio (%)	53.50	60.00	61.60	55.70	55.70	57.30

This is the event that we analyze. We assume that we invest in the BIST100 index at the end of the day in which this event occurred. In this example, we open our position when the daily index return is more than 5%.

On average, this event occurred every 47 days (approximately every month and a half) in the past. So if you believe that history repeats itself, then you may expect to see such an event in every 1.5 months.

This event occurred many times in the past. We calculate the returns as if we had invested in the BIST100 index when the event occurred and held our position for 1 week. This is the median value for those returns. The median is a measure of the central tendency and is typically more appropriate than the mean for cases with potential outlier values. Thus our calculations show that a typical return from investing in the event for a 1 week period is 2.06%

This is the historical proportion of cases where a positive return would have been achieved in all occurrences, and accordingly represents the historical probability of profiting from the event. For instance, a positive return was achieved in 57.30% of cases where we had invested in the BIST100 index at the time the event occurred, and had held our positions for 3 months.

Our benchmark

Statistics	Holding Period					
	1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
Median Return (%)	0.08	0.38	0.71	1.19	2.13	6.99
Hit Ratio (%)	51.5	54.0	55.8	56.7	57.5	64.2

These are the statistics calculated **using all available data**. For instance, among all 2 week periods in the historical period, 56.7% of them exhibited positive returns and a typical 2 week return of 1.19%.

We will use these statistics as a **benchmark** in our event profitability analysis.

In the following slides, events with:

- Median returns that are **3 times** higher than the benchmark median returns
 - Hit ratios that are higher than **66%**
 - An average frequency of occurrence less than **365 days**
- are highlighted in blue.

Summary of findings from event profitability analysis

- **There are no easy pickings:** Most hit ratios are only just above the benchmark
- Some events occurred **very infrequently** in the past. For instance, there are some events in the following tables that occurred only once or twice throughout the 24 year period that we analyzed. Even if such infrequent events offer promising profitability metrics, the sample size for such events is too small, so historical profitability metrics may not be a reliable indicator of future profitability.
- There are a small number of events where there is a reasonable frequency of occurrence and **promising historical profitability metrics**. These events are highlighted in the following tables.
- The profitability metrics presented here are typically **historical averages from many similar events** that occurred in the past. Thus, even if the future profitability of such events are similar to the historical patterns, if you trade such events only a **small number of times**, your resulting profit may significantly deviate from the typical returns, just because of what is known as 'sampling error'.
- For most cases, high hit ratios and positive median returns typically occur **after bullish events**, as opposed to bearish ones. This implies a momentum effect and favours being a 'trend follower' as opposed to being a 'contrarian' in the BIST.

Large returns

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
Daily return > 5%	47	Median Return (%)	0.36	1.72	2.06	2.59	2.50	5.26
		Hit Ratio (%)	53.50	60.00	61.60	55.70	55.70	57.30
Daily return > 7%	118	Median Return (%)	-0.21	1.69	2.06	2.81	2.89	4.58
		Hit Ratio (%)	45.90	60.80	64.90	52.70	55.40	56.80
Daily return > 10%	437	Median Return (%)	-1.86	0.77	1.30	4.24	4.93	0.80
		Hit Ratio (%)	35.00	55.00	60.00	65.00	60.00	50.00
Daily return < -5%	56	Median Return (%)	-0.04	-0.30	1.93	1.92	2.05	4.61
		Hit Ratio (%)	45.20	47.70	54.80	57.40	56.10	60.00
Daily return < -7%	159	Median Return (%)	0.98	0.78	1.46	1.21	1.90	6.83
		Hit Ratio (%)	52.70	52.70	54.50	52.70	52.70	61.80
Daily return < -10%	728	Median Return (%)	1.20	0.21	2.10	-0.25	-2.75	14.24
		Hit Ratio (%)	58.30	50.00	58.30	41.70	41.70	83.30

Source: Finansinvest

Large returns

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
Weekly return > 10%	24	Median Return (%)	0.21	0.94	1.82	2.45	2.95	10.54
		Hit Ratio (%)	51.70	56.40	56.70	58.10	58.40	64.00
Weekly return > 15%	68	Median Return (%)	-0.33	0.00	1.35	0.28	-0.14	6.23
		Hit Ratio (%)	44.20	49.60	54.30	51.20	49.60	58.10
Weekly return > 20%	190	Median Return (%)	-0.58	0.21	1.11	0.29	-2.07	-3.30
		Hit Ratio (%)	43.50	50.00	54.30	50.00	45.70	45.70
Weekly return < -10%	45	Median Return (%)	0.07	1.78	2.39	1.83	3.97	5.96
		Hit Ratio (%)	50.30	56.90	59.50	56.90	57.40	59.00
Weekly return < -15%	130	Median Return (%)	-0.32	0.69	4.31	1.55	1.80	11.35
		Hit Ratio (%)	46.30	52.20	64.20	50.70	55.20	68.70
Weekly return < 20%	546	Median Return (%)	1.17	6.27	6.17	-1.36	1.56	11.43
		Hit Ratio (%)	50.00	68.80	56.20	43.80	56.20	75.00

Source: Finansinvest

Large returns

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
Monthly return > 20%	15	Median Return (%)	0.21	0.72	0.97	2.50	3.07	11.70
		Hit Ratio (%)	51.90	53.90	54.90	56.20	57.20	67.10
Monthly return > 30%	33	Median Return (%)	0.04	0.81	1.02	0.10	-0.96	4.79
		Hit Ratio (%)	50.40	52.70	53.40	50.00	48.10	57.60
Monthly return > 40%	76	Median Return (%)	0.00	-0.04	0.00	-0.36	-3.20	1.06
		Hit Ratio (%)	47.80	47.80	49.60	46.10	42.60	53.00
Monthly return < -20%	60	Median Return (%)	0.80	0.09	0.69	1.69	3.28	0.39
		Hit Ratio (%)	55.90	51.00	54.50	55.20	55.90	51.70
Monthly return < -30%	265	Median Return (%)	0.66	3.61	6.54	0.36	-4.38	3.57
		Hit Ratio (%)	54.50	66.70	69.70	51.50	42.40	60.60
Monthly return < -40%	1457	Median Return (%)	3.60	23.36	29.05	11.79	1.04	9.09
		Hit Ratio (%)	50.00	83.30	100.00	83.30	66.70	100.00

Source: Finansinvest

Extreme returns

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
Daily return > 2 x Daily volatility	53	Median Return (%)	0.21	1.21	1.54	3.26	3.88	9.88
		Hit Ratio (%)	56.00	63.30	71.70	65.70	63.90	73.50
Daily return > 2.5 x Daily volatility	139	Median Return (%)	0.34	1.63	1.77	3.80	3.69	6.84
		Hit Ratio (%)	63.50	66.70	71.40	63.50	60.30	74.60
Daily return > 3 x Daily volatility	546	Median Return (%)	0.27	2.28	3.59	3.35	6.80	3.08
		Hit Ratio (%)	62.50	75.00	81.20	68.80	68.80	62.50
Daily return < -2 x Daily volatility	67	Median Return (%)	0.00	-0.23	0.82	0.96	0.81	3.93
		Hit Ratio (%)	49.60	47.30	55.00	56.50	54.20	60.30
Daily return < -2.5 x Daily volatility	178	Median Return (%)	0.00	-0.96	1.21	1.21	1.90	4.93
		Hit Ratio (%)	49.00	46.90	55.10	59.20	61.20	63.30
Daily return < -3 x Daily volatility	486	Median Return (%)	1.16	-1.22	0.86	0.99	-0.36	6.99
		Hit Ratio (%)	66.70	44.40	50.00	55.60	44.40	66.70

Source: Finansinvest

Consecutive return series with same sign

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
Positive returns in last 7 days	301	Median Return (%)	0.69	1.72	3.16	5.53	9.46	12.79
		Hit Ratio (%)	72.40	75.90	72.40	75.90	75.90	65.50
Positive returns in last 8 days	416	Median Return (%)	0.61	0.89	1.89	4.02	5.24	10.88
		Hit Ratio (%)	71.40	66.70	76.20	81.00	71.40	71.40
Positive returns in last 9 days	624	Median Return (%)	0.39	0.41	1.33	3.42	8.71	16.69
		Hit Ratio (%)	71.40	57.10	71.40	78.60	85.70	85.70
Positive returns in last 10 days	874	Median Return (%)	-1.07	0.08	1.02	3.33	11.34	23.09
		Hit Ratio (%)	10.00	50.00	70.00	60.00	70.00	80.00
Negative returns in last 7days	583	Median Return (%)	1.30	2.31	3.12	1.92	8.43	16.64
		Hit Ratio (%)	66.70	60.00	73.30	66.70	60.00	73.30
Negativereturns in last 8 days	2185	Median Return (%)	5.82	-0.60	3.62	-1.95	-8.53	4.26
		Hit Ratio (%)	50.00	50.00	50.00	25.00	25.00	50.00
Negativereturns in last 9 days	8741	Median Return (%)	-4.43	-6.33	0.63	3.16	-1.90	41.14
		Hit Ratio (%)	0.00	0.00	100.00	100.00	0.00	100.00
Negativereturns in last 10 days	8741	Median Return (%)	-5.96	0.00	13.25	3.97	3.97	47.68
		Hit Ratio (%)	0.00	0.00	100.00	100.00	100.00	100.00

Source: Finansinvest

Drawdowns

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
10% Drawdown	72	Median Return (%)	0.00	-0.03	0.20	0.20	1.16	5.36
		Hit Ratio (%)	49.60	49.60	52.10	51.20	53.70	59.50
15% Drawdown	78	Median Return (%)	0.19	0.51	1.29	2.36	4.41	3.89
		Hit Ratio (%)	54.50	52.70	57.10	56.20	63.40	60.70
20% Drawdown	123	Median Return (%)	0.18	0.00	1.27	2.35	2.92	4.18
		Hit Ratio (%)	53.50	49.30	59.20	67.60	64.80	66.20
No 10% or more drawdown since 2 months	546	Median Return (%)	0.06	1.05	1.16	0.47	2.86	15.23
		Hit Ratio (%)	50.00	62.50	62.50	56.20	56.20	75.00
No 10% or more drawdown since 3 months	971	Median Return (%)	0.63	1.33	3.39	7.92	9.58	17.26
		Hit Ratio (%)	66.70	88.90	88.90	88.90	88.90	88.90
No 20% or more drawdown since 1 month	380	Median Return (%)	-0.17	1.54	2.31	2.44	4.19	9.18
		Hit Ratio (%)	34.80	65.20	65.20	56.50	65.20	73.90
No 20% or more drawdown since 2 months	514	Median Return (%)	0.88	0.29	3.42	2.32	2.60	4.32
		Hit Ratio (%)	64.70	58.80	58.80	76.50	52.90	82.40
No 20% or more drawdown since 3 months	583	Median Return (%)	0.12	0.11	0.00	0.81	1.21	7.58
		Hit Ratio (%)	53.30	53.30	46.70	53.30	53.30	60.00
No 30% or more drawdown since 1 month	672	Median Return (%)	0.00	-0.35	-1.24	0.00	2.56	4.88
		Hit Ratio (%)	46.20	38.50	38.50	46.20	61.50	69.20
No 30% or more drawdown since 2 months	795	Median Return (%)	0.00	0.08	2.35	3.54	2.29	8.11
		Hit Ratio (%)	45.50	54.50	54.50	63.60	81.80	72.70
No 30% or more drawdown since 3 months	874	Median Return (%)	-1.42	-0.32	-0.59	0.20	-3.15	4.51
		Hit Ratio (%)	30.00	30.00	30.00	50.00	30.00	60.00

Source: Finansinvest

Deviation from moving averages

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
+10% deviation from 20 day moving average	87	Median Return (%)	0.29	1.13	1.93	4.26	7.91	16.02
		Hit Ratio (%)	56.00	65.00	62.00	63.00	68.00	71.00
+20% deviation from 20 day moving average	364	Median Return (%)	0.13	2.02	1.51	0.00	0.51	2.23
		Hit Ratio (%)	54.20	66.70	54.20	45.80	54.20	58.30
+30% deviation from 20 day moving average	971	Median Return (%)	0.04	0.53	2.87	3.33	-3.20	-1.36
		Hit Ratio (%)	55.60	55.60	66.70	55.60	44.40	44.40
-10% deviation from 20 day moving average	123	Median Return (%)	1.27	1.80	2.30	2.32	8.05	7.05
		Hit Ratio (%)	57.70	60.60	60.60	63.40	70.40	62.00
-20% deviation from 20 day moving average	1093	Median Return (%)	-2.29	1.80	0.70	1.54	-4.80	-4.29
		Hit Ratio (%)	37.50	50.00	50.00	50.00	37.50	37.50
-30% deviation from 20 day moving average	8741	Median Return (%)	-1.42	16.31	35.46	19.86	-2.84	11.35
		Hit Ratio (%)	0.00	100.00	100.00	100.00	0.00	100.00
+10% deviation from 50 day moving average	66	Median Return (%)	0.21	0.42	1.47	1.93	2.50	12.93
		Hit Ratio (%)	52.60	58.60	61.70	61.70	57.90	73.70
+20% deviation from 50 day moving average	162	Median Return (%)	0.11	2.53	3.71	5.16	7.89	21.56
		Hit Ratio (%)	50.00	68.50	66.70	68.50	75.90	72.20
+30% deviation from 50 day moving average	257	Median Return (%)	0.59	0.89	2.64	6.39	4.57	11.39
		Hit Ratio (%)	61.80	58.80	67.60	67.60	61.80	73.50

Source: Finansinvest

Deviation from moving averages

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
-10% deviation from 50 day moving average	111	Median Return (%)	-0.11	0.84	0.78	2.25	1.90	1.44
		Hit Ratio (%)	40.50	53.20	55.70	60.80	58.20	55.70
-20% deviation from 50 day moving average	350	Median Return (%)	0.34	2.81	4.31	0.27	4.16	-5.25
		Hit Ratio (%)	52.00	56.00	60.00	52.00	56.00	40.00
-30% deviation from 50 day moving average	1457	Median Return (%)	-0.85	-1.48	2.30	-4.63	5.19	5.76
		Hit Ratio (%)	16.70	50.00	50.00	33.30	66.70	66.70
+10% deviation from 200 day moving average	109	Median Return (%)	0.22	0.60	0.55	1.81	1.73	5.09
		Hit Ratio (%)	57.50	62.50	56.20	63.70	57.50	57.50
+20% deviation from 200 day moving average	114	Median Return (%)	0.12	0.70	1.22	2.80	4.21	6.22
		Hit Ratio (%)	51.90	61.00	61.00	62.30	62.30	59.70
+30% deviation from 200 day moving average	132	Median Return (%)	0.24	0.84	0.72	0.96	3.32	10.34
		Hit Ratio (%)	54.50	57.60	56.10	57.60	60.60	66.70
-10% deviation from 200 day moving average	186	Median Return (%)	0.00	0.00	-0.73	0.73	0.03	6.81
		Hit Ratio (%)	48.90	46.80	46.80	55.30	51.10	61.70
-20% deviation from 200 day moving average	364	Median Return (%)	0.29	0.40	0.85	-0.71	1.31	-0.94
		Hit Ratio (%)	54.20	54.20	50.00	45.80	54.20	45.80
-30% deviation from 200 day moving average	624	Median Return (%)	-0.52	3.05	-3.21	0.25	-3.36	18.87
		Hit Ratio (%)	42.90	57.10	42.90	50.00	35.70	57.10

Source: Finansinvest

Deviation from moving averages

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
+2 Sigma deviation from 20 day moving average	45	Median Return (%)	0.24	1.15	1.77	2.48	3.47	7.80
		Hit Ratio (%)	56.20	66.50	66.50	63.40	62.40	68.60
+2.5 Sigma deviation from 20 day moving average	130	Median Return (%)	0.16	1.21	2.22	4.26	4.76	10.42
		Hit Ratio (%)	53.70	67.20	73.10	65.70	70.10	65.70
+3 Sigma deviation from 20 day moving average	2185	Median Return (%)	-0.82	0.28	2.68	4.97	5.19	39.71
		Hit Ratio (%)	25.00	75.00	75.00	75.00	75.00	75.00
-2 Sigma deviation from 20 day moving average	59	Median Return (%)	0.09	0.35	0.83	1.07	2.76	5.16
		Hit Ratio (%)	51.40	54.10	57.40	56.80	56.80	62.20
-2.5 Sigma deviation from 20 day moving average	156	Median Return (%)	0.82	-0.22	-0.13	2.46	2.09	2.47
		Hit Ratio (%)	55.40	44.60	50.00	60.70	55.40	55.40
-3 Sigma deviation from 20 day moving average	672	Median Return (%)	1.00	-1.70	-0.76	2.76	1.90	2.71
		Hit Ratio (%)	61.50	15.40	38.50	69.20	53.80	53.80
+2 Sigma deviation from 50 day moving average	53	Median Return (%)	0.24	0.71	0.83	1.20	2.45	9.15
		Hit Ratio (%)	55.80	64.20	61.80	58.20	60.60	73.30
+2.5 Sigma deviation from 50 day moving average	117	Median Return (%)	0.34	1.41	1.82	4.14	6.67	11.27
		Hit Ratio (%)	60.00	66.70	65.30	69.30	69.30	80.00
+3 Sigma deviation from 50 day moving average	380	Median Return (%)	0.20	1.72	1.35	4.36	6.52	10.99
		Hit Ratio (%)	65.20	69.60	78.30	60.90	69.60	78.30

Source: Finansinvest

Deviation from moving averages

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
-2 Sigma deviation from 50 day moving average	114	Median Return (%)	-0.04	-0.15	0.01	0.84	1.02	5.75
		Hit Ratio (%)	42.90	48.10	50.60	53.20	55.80	61.00
-2.5 Sigma deviation from 50 day moving average	186	Median Return (%)	0.59	0.81	0.66	2.55	5.76	4.05
		Hit Ratio (%)	55.30	53.20	53.20	61.70	61.70	59.60
-3 Sigma deviation from 50 day moving average	624	Median Return (%)	2.18	2.53	5.18	6.13	5.01	0.77
		Hit Ratio (%)	64.30	57.10	64.30	64.30	64.30	50.00
+2 Sigma deviation from 200 day moving average	104	Median Return (%)	0.45	0.83	1.43	2.97	2.21	5.83
		Hit Ratio (%)	60.70	66.70	69.00	67.90	57.10	66.70
+2.5 Sigma deviation from 200 day moving average	148	Median Return (%)	0.16	1.28	1.06	2.89	1.14	11.94
		Hit Ratio (%)	55.90	67.80	67.80	57.60	54.20	72.90
+3 Sigma deviation from 200 day moving average	364	Median Return (%)	1.15	1.84	5.50	5.59	11.72	30.62
		Hit Ratio (%)	62.50	79.20	66.70	75.00	79.20	91.70
-2 Sigma deviation from 200 day moving average	336	Median Return (%)	0.19	2.34	3.86	5.73	5.17	-1.46
		Hit Ratio (%)	50.00	73.10	76.90	65.40	65.40	46.20
-2.5 Sigma deviation from 200 day moving average	971	Median Return (%)	-2.29	-2.12	0.38	1.19	4.16	-0.23
		Hit Ratio (%)	22.20	44.40	55.60	55.60	55.60	44.40
-3 Sigma deviation from 200 day moving average	1748	Median Return (%)	2.15	-0.95	1.97	6.83	-1.83	0.39
		Hit Ratio (%)	60.00	40.00	60.00	80.00	40.00	60.00

Source: Finansinvest

Moving average crossovers

Event	Average Frequency of Occurrence (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
5 day MA > 20 day MA	53	Median Return (%)	0.29	0.30	1.03	1.86	3.18	6.27
		Hit Ratio (%)	53.90	52.10	57.00	61.20	63.00	60.60
5 day MA < 20 day MA	53	Median Return (%)	-0.01	-0.33	-0.04	0.23	1.76	7.57
		Hit Ratio (%)	45.20	44.00	47.60	51.80	53.00	59.00
5 day MA > 50 day MA	90	Median Return (%)	0.00	0.44	1.26	2.83	3.16	8.33
		Hit Ratio (%)	47.40	54.60	63.90	61.90	66.00	66.00
5 day MA < 50 day MA	89	Median Return (%)	0.08	0.41	0.81	1.47	1.35	4.74
		Hit Ratio (%)	54.10	57.10	57.10	54.10	55.10	61.20
20 day MA > 50 day MA	146	Median Return (%)	0.26	0.73	0.99	2.50	3.99	5.66
		Hit Ratio (%)	65.00	61.70	58.30	58.30	61.70	68.30
20 day MA < 50 day MA	143	Median Return (%)	0.05	0.40	1.04	-0.83	-2.19	6.70
		Hit Ratio (%)	50.80	55.70	59.00	45.90	47.50	67.20
5 day MA > 100 day MA	165	Median Return (%)	0.13	0.64	1.09	1.55	0.84	7.60
		Hit Ratio (%)	54.70	66.00	60.40	64.20	52.80	67.90
5 day MA < 100 day MA	165	Median Return (%)	-0.56	1.13	-0.18	1.06	-0.43	4.84
		Hit Ratio (%)	47.20	54.70	49.10	54.70	47.20	58.50
20 day MA > 100 day MA	301	Median Return (%)	-0.21	0.31	0.34	0.73	2.48	12.07
		Hit Ratio (%)	48.30	51.70	55.20	51.70	58.60	72.40
20 day MA < 100 day MA	301	Median Return (%)	0.00	-0.09	-0.39	-0.90	-2.83	2.56
		Hit Ratio (%)	48.30	48.30	44.80	44.80	34.50	58.60

Source: Finansinvest

Moving average crossovers

Event	Average Frequency of Occurrence (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
50 day MA > 100 day MA	350	Median Return (%)	0.20	1.82	2.38	2.79	3.55	13.11
		Hit Ratio (%)	56.00	64.00	68.00	68.00	68.00	68.00
50 day MA < 100 day MA	336	Median Return (%)	0.22	1.10	2.31	1.22	1.26	8.67
		Hit Ratio (%)	50.00	61.50	61.50	57.70	61.50	69.20
5 day MA > 200 day MA	291	Median Return (%)	-0.06	1.73	0.76	1.69	0.79	8.07
		Hit Ratio (%)	46.70	63.30	63.30	56.70	53.30	66.70
5 day MA < 200 day MA	273	Median Return (%)	0.00	0.77	1.43	0.21	0.03	3.73
		Hit Ratio (%)	43.80	62.50	59.40	53.10	50.00	68.80
20 day MA > 200 day MA	486	Median Return (%)	-0.08	0.91	1.33	2.26	3.14	10.28
		Hit Ratio (%)	33.30	61.10	72.20	61.10	55.60	66.70
20 day MA < 200 day MA	460	Median Return (%)	0.32	-1.39	-1.45	0.53	2.64	2.56
		Hit Ratio (%)	57.90	31.60	47.40	57.90	63.20	68.40
50 day MA > 200 day MA	583	Median Return (%)	-0.14	0.44	0.78	1.68	6.41	19.79
		Hit Ratio (%)	46.70	53.30	53.30	66.70	73.30	73.30
50 day MA < 200 day MA	546	Median Return (%)	-0.11	1.30	2.22	2.72	7.11	10.00
		Hit Ratio (%)	43.80	62.50	75.00	81.20	68.80	87.50
100 day MA > 200 day MA	728	Median Return (%)	-0.07	2.73	3.99	3.80	6.38	14.79
		Hit Ratio (%)	41.70	66.70	75.00	83.30	75.00	91.70
100 day MA < 200 day MA	728	Median Return (%)	-0.01	1.09	0.95	6.53	13.26	12.75
		Hit Ratio (%)	50.00	58.30	66.70	75.00	75.00	66.70

Source: Finansinvest

Deviation from trend lines

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
+5% deviation from 20 day linear trend line	49	Median Return (%)	0.00	1.47	1.86	3.03	3.18	8.50
		Hit Ratio (%)	49.70	58.80	59.90	60.50	61.60	63.30
-5% deviation from 20 day linear trend line	46	Median Return (%)	0.00	0.34	1.00	1.02	1.22	7.52
		Hit Ratio (%)	47.60	51.80	56.00	55.50	52.90	62.80
+10% deviation from 20 day linear trend line	257	Median Return (%)	-0.17	-0.49	0.42	-1.58	-0.60	1.34
		Hit Ratio (%)	44.10	44.10	52.90	47.10	47.10	50.00
-10% deviation from 20 day linear trend line	350	Median Return (%)	1.66	4.01	6.80	6.85	8.43	7.96
		Hit Ratio (%)	72.00	60.00	76.00	64.00	68.00	68.00
+15% deviation from 20 day linear trend line	874	Median Return (%)	-1.74	-1.39	-2.43	-7.34	-6.07	-1.57
		Hit Ratio (%)	50.00	50.00	40.00	20.00	20.00	40.00
-15% deviation from 20 day linear trend line	2914	Median Return (%)	-4.64	-12.89	-7.49	-18.04	-15.46	11.86
		Hit Ratio (%)	0.00	33.30	33.30	0.00	33.30	66.70
+5% deviation from 65 day linear trend line	51	Median Return (%)	0.17	0.37	1.39	2.70	2.25	6.47
		Hit Ratio (%)	53.50	55.30	62.90	61.20	58.80	61.20
-5% deviation from 65 day linear trend line	43	Median Return (%)	-0.15	-0.41	0.04	0.57	2.70	7.79
		Hit Ratio (%)	46.00	47.50	50.00	53.00	57.40	63.40
+10% deviation from 65 day linear trend line	82	Median Return (%)	1.04	2.13	3.87	5.07	5.94	11.55
		Hit Ratio (%)	58.50	65.10	67.90	66.00	65.10	69.80

Source: Finansinvest

Deviation from trend lines

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
-10% deviation from 65 day linear trend line	87	Median Return (%)	0.13	0.81	0.96	2.62	3.32	6.06
		Hit Ratio (%)	51.00	57.00	57.00	64.00	59.00	62.00
+15% deviation from 65 day linear trend line	159	Median Return (%)	0.27	0.92	1.57	0.00	2.04	8.87
		Hit Ratio (%)	52.70	60.00	61.80	49.10	52.70	63.60
-15% deviation from 65 day linear trend line	213	Median Return (%)	0.00	2.10	1.22	1.83	6.05	1.53
		Hit Ratio (%)	48.80	53.70	56.10	58.50	61.00	51.20
+10% deviation from 260 day linear trend line	96	Median Return (%)	0.27	0.77	0.90	1.07	2.43	6.47
		Hit Ratio (%)	53.80	58.20	53.80	53.80	56.00	70.30
-10% deviation from 260 day linear trend line	120	Median Return (%)	0.19	0.10	0.63	2.69	3.97	3.86
		Hit Ratio (%)	56.20	50.70	57.50	60.30	61.60	58.90
+20% deviation from 260 day linear trend line	208	Median Return (%)	1.55	2.41	3.74	6.05	8.97	11.29
		Hit Ratio (%)	73.80	71.40	73.80	69.00	66.70	71.40
-20% deviation from 260 day linear trend line	186	Median Return (%)	-0.98	-0.24	1.93	0.78	4.41	0.39
		Hit Ratio (%)	40.40	46.80	63.80	55.30	63.80	53.20
+30% deviation from 260 day linear trend line	257	Median Return (%)	-0.09	1.44	2.93	-1.10	3.73	6.34
		Hit Ratio (%)	41.20	55.90	58.80	44.10	55.90	58.80
-30% deviation from 260 day linear trend line	728	Median Return (%)	-1.22	2.83	0.60	2.42	-3.92	8.91
		Hit Ratio (%)	25.00	66.70	58.30	75.00	33.30	66.70

Source: Finansinvest

Deviation from trend lines

Event	Average Frequency of Occurrence (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
+1.5 sigma deviation from 20 day linear trend line	62	Median Return (%)	0.14	0.97	2.25	4.08	4.70	10.24
		Hit Ratio (%)	51.80	61.70	67.40	68.80	68.80	64.50
-1.5 sigma deviation from 20 day linear trend line	59	Median Return (%)	-0.26	-0.73	0.24	0.23	0.93	3.18
		Hit Ratio (%)	44.90	42.90	51.70	51.00	53.10	55.10
+2 sigma deviation from 20 day linear trend line	203	Median Return (%)	0.27	1.01	3.28	5.08	5.26	15.03
		Hit Ratio (%)	55.80	62.80	72.10	79.10	74.40	76.70
-2 sigma deviation from 20 day linear trend line	141	Median Return (%)	0.00	-0.21	-0.42	-0.01	1.41	4.71
		Hit Ratio (%)	46.80	43.50	45.20	50.00	56.50	62.90
+2.5 sigma deviation from 20 day linear trend line	1093	Median Return (%)	0.30	2.18	2.19	6.60	15.21	33.73
		Hit Ratio (%)	62.50	75.00	75.00	62.50	75.00	87.50
-2.5 sigma deviation from 20 day linear trend line	583	Median Return (%)	0.58	-0.34	-0.46	2.66	2.28	9.44
		Hit Ratio (%)	53.30	40.00	46.70	66.70	60.00	73.30
+1.5 sigma deviation from 65 day linear trend line	86	Median Return (%)	0.18	1.10	1.84	2.24	4.21	10.48
		Hit Ratio (%)	56.90	65.70	64.70	59.80	61.80	70.60
-1.5 sigma deviation from 65 day linear trend line	79	Median Return (%)	0.02	0.57	1.30	1.68	2.76	2.82
		Hit Ratio (%)	50.00	54.50	58.20	59.10	57.30	57.30
+2 sigma deviation from 65 day linear trend line	86	Median Return (%)	0.18	1.10	1.84	2.24	4.21	10.48
		Hit Ratio (%)	56.90	65.70	64.70	59.80	61.80	70.60

Source: Finansinvest

Deviation from trend lines

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
-2 sigma deviation from 65 day linear trend line	79	Median Return (%)	0.02	0.57	1.30	1.68	2.76	2.82
		Hit Ratio (%)	50.00	54.50	58.20	59.10	57.30	57.30
+2.5 sigma deviation from 65 day linear trend line	460	Median Return (%)	0.01	0.00	0.34	1.64	2.04	-0.34
		Hit Ratio (%)	52.60	47.40	52.60	57.90	57.90	47.40
-2.5 sigma deviation from 65 day linear trend line	486	Median Return (%)	1.15	0.04	2.53	2.66	0.35	-1.28
		Hit Ratio (%)	61.10	50.00	61.10	72.20	55.60	50.00
+1.5 sigma deviation from 260 day linear trend line	137	Median Return (%)	0.14	0.44	0.63	2.95	4.11	8.08
		Hit Ratio (%)	54.70	54.70	56.20	59.40	64.10	65.60
-1.5 sigma deviation from 260 day linear trend line	219	Median Return (%)	0.16	1.00	2.89	3.30	2.49	-1.19
		Hit Ratio (%)	52.50	60.00	67.50	62.50	65.00	45.00
+2 sigma deviation from 260 day linear trend line	291	Median Return (%)	0.57	1.83	5.11	5.31	8.54	7.69
		Hit Ratio (%)	60.00	63.30	70.00	73.30	76.70	80.00
-2 sigma deviation from 260 day linear trend line	546	Median Return (%)	-0.06	0.75	1.56	0.33	6.63	3.18
		Hit Ratio (%)	43.80	56.20	56.20	56.20	68.80	50.00
+2.5 sigma deviation from 260 day linear trend line	437	Median Return (%)	-1.18	-0.73	-1.95	1.14	0.80	0.59
		Hit Ratio (%)	35.00	45.00	40.00	50.00	55.00	50.00
-2.5 sigma deviation from 260 day linear trend line	874	Median Return (%)	2.38	1.82	0.01	1.87	4.45	0.97
		Hit Ratio (%)	80.00	60.00	50.00	60.00	70.00	60.00

Source: Finansinvest

RSI

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
RSI > 70	54	Median Return (%)	0.42	0.70	1.17	1.21	2.54	10.86
		Hit Ratio (%)	60.20	59.00	60.20	60.20	62.10	77.60
RSI < 70	54	Median Return (%)	0.29	0.73	0.62	1.41	2.40	9.78
		Hit Ratio (%)	55.60	59.30	59.90	58.60	60.50	71.60
RSI > 30	151	Median Return (%)	0.00	-0.72	0.42	0.49	2.46	1.84
		Hit Ratio (%)	44.80	43.10	55.20	51.70	60.30	55.20
RSI < 30	151	Median Return (%)	-0.09	-0.28	0.56	2.66	3.73	2.76
		Hit Ratio (%)	43.10	48.30	53.40	56.90	60.30	60.30

Source: Finansinvest

Breakouts

Event	Average Frequency of Occurrence (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
Index increased to a new high	19	Median Return (%)	0.43	1.37	1.88	3.76	4.94	11.94
		Hit Ratio (%)	59.90	66.70	67.80	69.10	64.20	77.00
20 day range breakout up	8	Median Return (%)	0.24	0.80	1.16	2.46	3.18	9.59
		Hit Ratio (%)	55.90	61.80	61.90	63.60	62.60	71.60
65 day range breakout up	12	Median Return (%)	0.29	0.86	1.17	2.56	3.49	10.59
		Hit Ratio (%)	57.70	62.20	63.80	65.20	63.20	76.20
5 day range breakout up	4	Median Return (%)	0.20	0.70	1.06	2.25	3.03	7.60
		Hit Ratio (%)	54.50	58.90	60.10	61.70	61.10	65.80
5 day range breakout down	6	Median Return (%)	0.00	0.00	0.49	0.75	1.74	7.06
		Hit Ratio (%)	49.20	49.80	53.00	54.20	55.50	63.70
20 day range breakout down	16	Median Return (%)	0.00	0.16	0.79	0.74	1.17	5.20
		Hit Ratio (%)	48.90	51.80	54.80	54.30	55.00	63.40
65 day range breakout down	42	Median Return (%)	0.00	0.00	0.42	0.77	3.50	3.93
		Hit Ratio (%)	49.00	48.60	52.40	53.80	62.40	62.40
Positive return largest in absolute terms since 20-days	52	Median Return (%)	0.27	0.89	1.48	2.54	3.69	7.90
		Hit Ratio (%)	56.30	59.90	67.10	59.90	59.90	69.50
Positive return largest in absolute terms since 5-days	13	Median Return (%)	0.29	1.07	1.49	2.24	3.06	7.40
		Hit Ratio (%)	56.60	60.70	63.20	60.40	60.60	67.20
Positive return largest in absolute terms since 65-days	159	Median Return (%)	0.19	2.33	2.69	2.83	3.49	4.82
		Hit Ratio (%)	52.70	70.90	70.90	60.00	56.40	67.30

Source: Finansinvest

Gaps

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
Opening up the gap (i.e. today's open is higher than yesterday's high)	15	Median Return (%)	0.03	0.29	0.43	0.75	0.99	4.33
		Hit Ratio (%)	51.50	56.10	55.30	56.60	54.20	61.10
Opening down gap (i.e. today's open is lower than yesterday's low)	29	Median Return (%)	0.00	0.18	0.49	0.03	-0.16	3.30
		Hit Ratio (%)	49.30	52.00	52.30	50.00	48.40	56.90
Full up gap (i.e. today's low is higher than yesterday's high)	39	Median Return (%)	0.00	0.28	0.81	1.10	0.91	5.00
		Hit Ratio (%)	47.30	55.00	58.60	58.10	53.60	63.10
Full down gap (i.e. today's high is lower than yesterday's low)	72	Median Return (%)	0.00	0.00	0.09	-0.13	-0.46	3.73
		Hit Ratio (%)	47.90	47.90	50.40	46.30	45.50	57.00

Source: Finansinvest

Volume breakouts

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
Daily return is positive and daily volume is more than twice the 3 month average volume	672	Median Return (%)	0.27	0.65	1.26	8.25	10.97	0.71
		Hit Ratio (%)	61.50	61.50	53.80	76.90	76.90	53.80
Daily return is negative and daily volume is more than twice the 3 month average volume	8741	Median Return (%)	-3.39	-6.09	2.84	2.11	-1.32	-12.21
		Hit Ratio (%)	0.00	0.00	100.00	100.00	0.00	0.00
Daily return is positive and daily volume is more than 2.5 times the 3 month average volume	2185	Median Return (%)	-0.21	-1.61	-1.23	1.01	3.05	-7.83
		Hit Ratio (%)	50.00	50.00	50.00	50.00	50.00	25.00
Daily return is negative and daily volume is more than 2.5 times the 3 month average volume	8741	Median Return (%)	-3.39	-6.09	2.84	2.11	-1.32	-12.21
		Hit Ratio (%)	0.00	0.00	100.00	100.00	0.00	0.00

Source: Finansinvest

Extreme volatility

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
Daily volatility < 10-percentile of daily volatilities observed in the last 12 months	79	Median Return (%)	0.16	0.02	0.43	0.84	0.63	8.95
		Hit Ratio (%)	57.70	50.50	51.40	56.80	51.40	74.80
Daily volatility > 10-percentile of daily volatilities observed in the last 12 months	79	Median Return (%)	-0.01	0.05	0.50	1.11	1.08	10.17
		Hit Ratio (%)	46.40	50.90	56.40	56.40	55.50	76.40
Daily volatility < 90-percentile of daily volatilities observed in the last 12 months	178	Median Return (%)	-0.06	0.97	1.82	1.31	2.87	7.12
		Hit Ratio (%)	46.90	59.20	55.10	57.10	59.20	63.30
Daily volatility > 90-percentile of daily volatilities observed in the last 12 months	168	Median Return (%)	0.01	0.90	2.29	1.28	1.09	6.23
		Hit Ratio (%)	50.00	61.50	61.50	53.80	55.80	59.60

Source: Finansinvest

Cross asset divergence

Event	Average Frequency of Occurance (Days)	Statistics	Holding Period					
			1 Day	3 Days	1 Week	2 Weeks	1 Month	3 Months
Daily BIST100 return < 0	25	Median Return (%)	0.28	0.53	0.62	1.40	2.53	3.65
Daily benchmark bond yield change <= 0		Hit Ratio (%)	57.1	56.0	56.5	58.8	59.4	59.4
Daily USD/TRY return <= 0								
Daily BIST100 return > 0	24	Median Return (%)	-0.19	-0.21	0.19	0.43	0.48	1.75
Daily benchmark bond yield change >= 0		Hit Ratio (%)	47.3	47.0	51.1	53.3	51.6	54.1
Daily USD/TRY return >= 0								
Weekly BIST100 return < 0	31	Median Return (%)	0.17	0.28	0.71	1.13	0.97	5.81
Weekly benchmark bond yield change <= 0		Hit Ratio (%)	54.3	52.5	54.3	58.3	54.7	58.3
Weekly USD/TRY return <= 0								
Weekly BIST100 return > 0	28	Median Return (%)	0.25	0.28	0.68	1.14	0.87	5.37
Weekly benchmark bond yield change >= 0		Hit Ratio (%)	54.7	53.1	56.7	58.0	52.1	57.3
Weekly USD/TRY return >= 0								

Source: Finansinvest

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